

Python Programming Questions with Solution Code

Ques – 1 - Write a function named dubs to find all duplicates in a list.

```
1  def dubs(lst):
2      duplicates = []
3      for item in lst:
4          if lst.count(item) > 1 and item not in duplicates:
5              duplicates.append(item)
6      return duplicates
7
8  numbers = [1, 2, 3, 2, 4, 5, 3, 6, 2]
9  print("Duplicate elements:", dubs(numbers))
10
```

Ques – 2 - By using a for loop, write a program that prints the decimal equivalent of 1/2, 1/3, 1/4, ... 1/10.

```
1  for i in range(2, 11):
2      print(f"1/{i} = {1 / i}")
```

Ques – 3 - a Python program to find the minimum and maximum element in a list.

```
1 numbers = [12, 5, 89, 34, 2, 67, 45]
2 minimum = min(numbers)
3 maximum = max(numbers)
4 print("Minimum:", minimum)
5 print("Maximum:", maximum)
```

Ques – 4 - Write a Python program to find the addition of two square matrices.

```
1 v A = [
2     [1, 2],
3     [3, 4]
4 ]
5 v B = [
6     [5, 6],
7     [7, 8]
8 ]
9 n = len(A)
10 result = [[0] * n for _ in range(n)]
11 v for i in range(n):
12 v     for j in range(n):
13         result[i][j] = A[i][j] + B[i][j]
14
15 print("Sum of matrices:")
16 v for row in result:
17     print(row)
```

Ques – 5 -Write a Python program to remove empty strings from a list of strings.

```
1  states = ["DL", "", "TN", "", "UP", "KK", ""]
2
3  result = [item for item in states if item != ""]
4
5  print("Original list:", states)
6  print("After removing empty strings:", result)
```

Ques – 6 - Write a function unique to find all the unique elements of a list.

```
1  def unique(lst):
2      result = []
3      for item in lst:
4          if item not in result:
5              result.append(item)
6      return result
7
8  numbers = [1, 2, 2, 3, 4, 4, 5, 1]
9  print("Unique elements:", unique(numbers))
```

Ques – 7 -Implement a Python script for factorial of a number by using recursion.

```
1 def factorial(n):
2     if n == 0 or n == 1:
3         return 1
4     return n * factorial(n - 1)
5
6 num = 5
7 print("Factorial:", factorial(num))
```

Ques – 8 - From the given text, perform the following tasks: (a) Find how many words are starting with a capital letter. (b) Create a dictionary of vowels as keys and count as values.

```
1 text = "Python is a popular Programming Language. It is used for Data Science, Web Development and Automation."
2
3 # (a) Count words beginning with a capital letter
4 # Split the text by spaces and strip punctuation from the words
5 words = [word.strip(",.!?\"'") for word in text.split()]
6 capital_words = [word for word in words if word and word[0].isupper()]
7
8 print("Words starting with a capital letter:", len(capital_words))
9 print("Capitalized words:", capital_words)
10
11 # (b) Count vowels
12 vowels = "aeiou"
13 vowel_count = {v: 0 for v in vowels}
14
15 for char in text.lower():
16     if char in vowel_count:
17         vowel_count[char] += 1
18
19 print("Vowel dictionary:", vowel_count)
```

Ques – 9 -Write a Python program to find the GCD of two numbers.

```
1  a = 48
2  b = 18
3
4  while b != 0:
5      a, b = b, a % b
6
7  print("GCD:", a)
```

Ques – 10 - Write a Python program to check whether a number is prime or not.

```
1  num = 29
2  if num < 2:
3      print("Not a prime number")
4  else:
5      is_prime = True
6
7  for i in range(2, int(num ** 0.5) + 1):
8      if num % i == 0:
9          is_prime = False
10         break
11
12  if is_prime:
13      print("Prime number")
14  else:
15      print("Not a prime number")
```

Ques – 11 -Write a Python program to print Mean, Mode and Median of a list or array.

```
1  from statistics import mean, median, mode
2
3  numbers = [2, 4, 4, 5, 6, 8, 4, 9]
4
5  print("Mean:", mean(numbers))
6  print("Median:", median(numbers))
7  print("Mode:", mode(numbers))
```

Ques – 12 -Write a Python program to check whether a number is an Armstrong number or not.

```
1  num = 153
2
3  digits = str(num)
4  power = len(digits)
5  total = sum(int(digit) ** power for digit in digits)
6
7  if total == num:
8      print("Armstrong number")
9  else:
10     print("Not an Armstrong number")
```

Ques – 13 - Write a Python program to print Fibonacci series of n numbers.

```
1  n = 10
2  a, b = 0, 1
3
4  print("Fibonacci series:")
5
6  for _ in range(n):
7      print(a, end=" ")
8      a, b = b, a + b
```

Ques – 14 -Write a Python program to check how many vowels are in a paragraph.

```
1  paragraph = """Python is easy to learn.
2  It is widely used in software development."""
3
4  vowels = "aeiouAEIOU"
5  count = 0
6
7  for char in paragraph:
8      if char in vowels:
9          count += 1
10
11  print("Number of vowels:", count)
```

Ques – 15 - Write a Python program to reverse:

(a) Digit of an integer (b) A list (c) A tuple (d) A string.

```
1  # (a) Reverse digits of an integer
2  num = 12345
3  reverse_num = int(str(num)[::-1])
4  print("Reversed integer:", reverse_num)
5
6  # (b) Reverse a list
7  numbers = [1, 2, 3, 4, 5]
8  print("Reversed list:", numbers[::-1])
9
10 # (c) Reverse a tuple
11 values = (10, 20, 30, 40)
12 print("Reversed tuple:", values[::-1])
13
14 # (d) Reverse a string
15 text = "Python"
16 print("Reversed string:", text[::-1])
```

Ques – 16 -Write a Python program to find GCD through the Euclidean method.

```
1  def gcd_euclidean(a, b):  
2      while b != 0:  
3          a, b = b, a % b  
4      return a  
5  
6  a = 48  
7  b = 18  
8  
9  print("GCD using Euclidean method:", gcd_euclidean(a, b))  
10
```

Ques – 17 -Write a Python program to convert decimal numbers to binary.

```
1  num = 25  
2  
3  binary = bin(num)[2:]  
4  
5  print("Decimal:", num)  
6  print("Binary:", binary)
```

Ques – 18 - Write a Python program to print the sum of prime numbers up to a given number.

```
1  n = 20
2  sum_primes = 0
3
4  for num in range(2, n + 1):
5      is_prime = True
6
7      for i in range(2, int(num ** 0.5) + 1):
8          if num % i == 0:
9              is_prime = False
10             break
11
12     if is_prime:
13         sum_primes += num
14
15     print("Sum of prime numbers up to", n, "=", sum_primes)
16
```

Ques – 19 -Write a Python program to find the sum of all elements in a list.

```
1  numbers = [10, 20, 30, 40, 50]
2
3  total = 0
4
5  for num in numbers:
6      total += num
7
8  print("Sum of elements:", total)
```

Ques – 20 -Write a Python program to count the frequency of each element in a list.

```
1  numbers = [1, 2, 2, 3, 1, 4, 2, 3]
2
3  frequency = {}
4
5  for num in numbers:
6      if num in frequency:
7          frequency[num] += 1
8      else:
9          frequency[num] = 1
10
11 print("Frequency of elements:", frequency)
```

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